

Guide 1

Logical Volumes and File Systems Management

2026

The main goal of this guide is to understand how to use and configure Logical Volumes (LVs) and File Systems (FSs) in the Linux environment. At the end, we will end up with a setup similar to the one described in Figure 1.

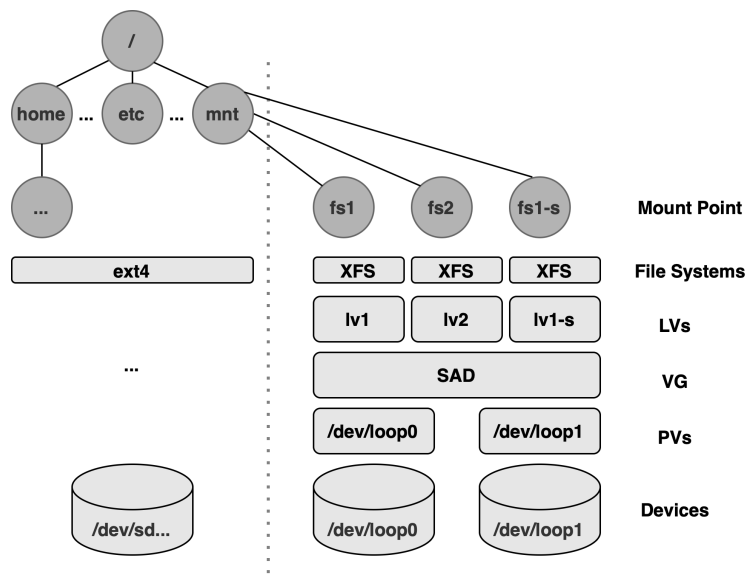


Figure 1: Final Setup of Guide1 with multiple devices and file systems using LVM.

1 Setup

For the exercises described next, use the VM from *Guide0*. Also, use the commands below to set up two loop disk devices, each with 2 GB. These are pseudo-devices that are backed by a file. We are creating these devices, instead of using real disks (or partitions), for ease of deployment. **Note**, however, that the Logical Volume Management (LVM) and file system tasks discussed in this guide would apply to real devices.

1. Use the `dd` command to create two files, each with 2 GB.

```
dd if=/dev/zero of=blockfile1 bs=1M count=2048
dd if=/dev/zero of=blockfile2 bs=1M count=2048
```

2. Next connect each file to a distinct loop disk device.

```
sudo losetup /dev/loop0 blockfile1
sudo losetup /dev/loop1 blockfile2
```

3. Check that the two disks (loop devices) `/dev/loop0` and `/dev/loop1` have been created successfully with the `fdisk -l` command.

2 Logical Volume Manager

Our next goal is to create two logical volumes (as an alternative to creating physical disk partitions) at the `/dev/loop0` device. For such a task, explore the commands below.

1. Create a Physical Volume (PV) for the device. Use the `pvsdisplay` command to check the created PV.

```
sudo pvcreate /dev/loop0
```

2. Create a Volume Group (VG), named `SAD`, that will include the `/dev/loop0` PV. Use the `vgdisplay` command to check the created VG. With the command `pvsdisplay` check that the `/dev/loop0` PV is now associated with the `SAD` VG.

```
sudo vgcreate SAD /dev/loop0
```

3. Create two Logical Volumes (`lv1` and `lv2`), each with 200 MiB. With the commands `lvdisplay`, check that the volumes were created.

```
sudo lvcreate --size 200M --name lv1 /dev/SAD
sudo lvcreate --size 200M --name lv2 /dev/SAD
```

4. Use the commands `fdisk -l` and `ls -la /dev/SAD` to check the two volumes mapped by the *Device Mapper*. The Device Mapper is a kernel driver that provides a framework for volume management. It provides a generic way of creating mapped devices, which may be used as logical volumes.

3 Filesystem

Now that we have two LVs, we want to initialize a distinct file system in each. Let us start with `lv1`.

1. Initialize a XFS filesystem (*i.e.*, its data and metadata) in `lv1`.

```
sudo mkfs.xfs /dev/SAD/lv1
```

2. Unfortunately, we have not allocated enough space for `lv1`. Extend the LV's size and initialize the filesystem again.

```
sudo lvextend -L+100M /dev/SAD/lv1
sudo mkfs.xfs /dev/SAD/lv1
```

3. Create a mount point for the file system (*i.e.*, its root folder). Then, mount the file system. The `mount` command tells the kernel to attach the filesystem found on a device (*e.g.*, `/dev/SAD/lv1`) at a given directory (*e.g.*, `/mnt/fs1`). Check if the file system was created with the `df -h` command.

```
sudo mkdir /mnt/fs1
sudo mount /dev/SAD/lv1 /mnt/fs1
```

4. Copy the folder `/etc` to the root folder of the new file system.

```
sudo cp -r /etc /mnt/fs1/
```

5. What if we want to set up another FS for `lv2`?
Reproduce the previous steps to attach a new file system, found in device `/dev/SAD/lv2`, at the directory `/mnt/fs2`.

4 Resizing the File System

Again, our math was wrong, and we need the file system to grow further. Let us expand our file system residing at `lv1` to use 400MB.

1. First, extend the logical volume to the correct size

```
sudo lvextend -L+100M /dev/SAD/lv1
```

2. Then, grow the file system. Check if the file system was extended with the `df -h` command.

```
sudo xfs_growfs /mnt/fs1
```

5 Extending Volume Groups and Migrating Logical Volumes

We noticed that our `/dev/loop0` device is failing, and we want to move its data to the second disk (`/dev/loop1`).

1. Create and add the second PV to the `SAD` Volume Group. Use the `vgdisplay` command to check the extended VG (i.e., observe the `Cur PV` and `Act PV` stats). With the command `pvdiskdisplay`, check that the PV is now associated with that VG.

```
sudo pvcreate /dev/loop1
sudo vgextend SAD /dev/loop1
```

2. Before migrating data across the PVs, *umount* "stop" the file systems to avoid any data and metadata corruption.

```
sudo umount /mnt/fs1
sudo umount /mnt/fs2
```

3. Now, migrate data across PVs, and then remove the faulty PV from the `SAD` VG.

```
sudo pvmove /dev/loop0 /dev/loop1
sudo vgreduce SAD /dev/loop0
```

4. Check that the logical volumes (`lv1` and `lv2`) still exist (`pvdiskdisplay`, `vgdisplay`, `lvdisplay`), mount the two file systems, and check their data.

6 Logical Volumes Snapshots

Finally, we may want to create a snapshot of a given LV for backup or versioning purposes (e.g., *lv1*).

1. Create a snapshot of *lv1*, named *lv1-s*. Use the `lvdisplay` command to check that the operation completed successfully. Also, look at the size of the snapshot (`Allocated to snapshot` field). The snapshot is not using any space because of the Copy-on-Write (CoW) technique.

```
sudo lvcreate -s /dev/SAD/lv1 -L 400M -n lv1-s
```

2. Create a new mount point (e.g., */mnt/fs1-s*) and mount the file system residing on */dev/SAD/lv1-s*. Look again at the size of the snapshot (`Allocated to snapshot` field). Now the snapshot is occupying a small percentage of space due to changes in the file system mounted on top of *lv1-s* (i.e., copies are made to store updated content at *lv1-s*).

```
sudo mkdir /mnt/fs1-s
sudo mount -o nouuid /dev/SAD/lv1-s /mnt/fs1-s
```

3. Each file system must have a unique identifier (*uuid*). However, both the file systems at *lv1* and *lv1-s* volumes share the same ID. Previously, we used the `-o nouuid` flag for mounting the new file system without getting an error. Another option is to change its *uuid*.

```
sudo umount /mnt/fs1-s
sudo xfs_admin -U generate /dev/SAD/lv1-s
```

4. Mount the volume again, without using the `-o nouuid` flag, and check the mount point's content. Look again at the size of the snapshot (`Allocated to snapshot` field). More changes to the file system (mounted on top of *lv1-s*) mean more space allocated.
5. Copy the folder */opt* to the root folder of the file system mounted on */mnt/fs1*. Look again at the size of the snapshot (`Allocated to snapshot` field). When the file system, mounted on top of *lv1*, is modified, the snapshot size also increases (i.e., copies are also made to store old content in the snapshot *lv1-s* before updating *lv1*).

```
sudo cp -r /opt /mnt/fs1/
```

Learning outcomes

Experiment Logical Volume Management (LVM) deployment and configuration. Experiment with file system deployment and management. Assess how LVM helps simplify the orchestration of storage resources.